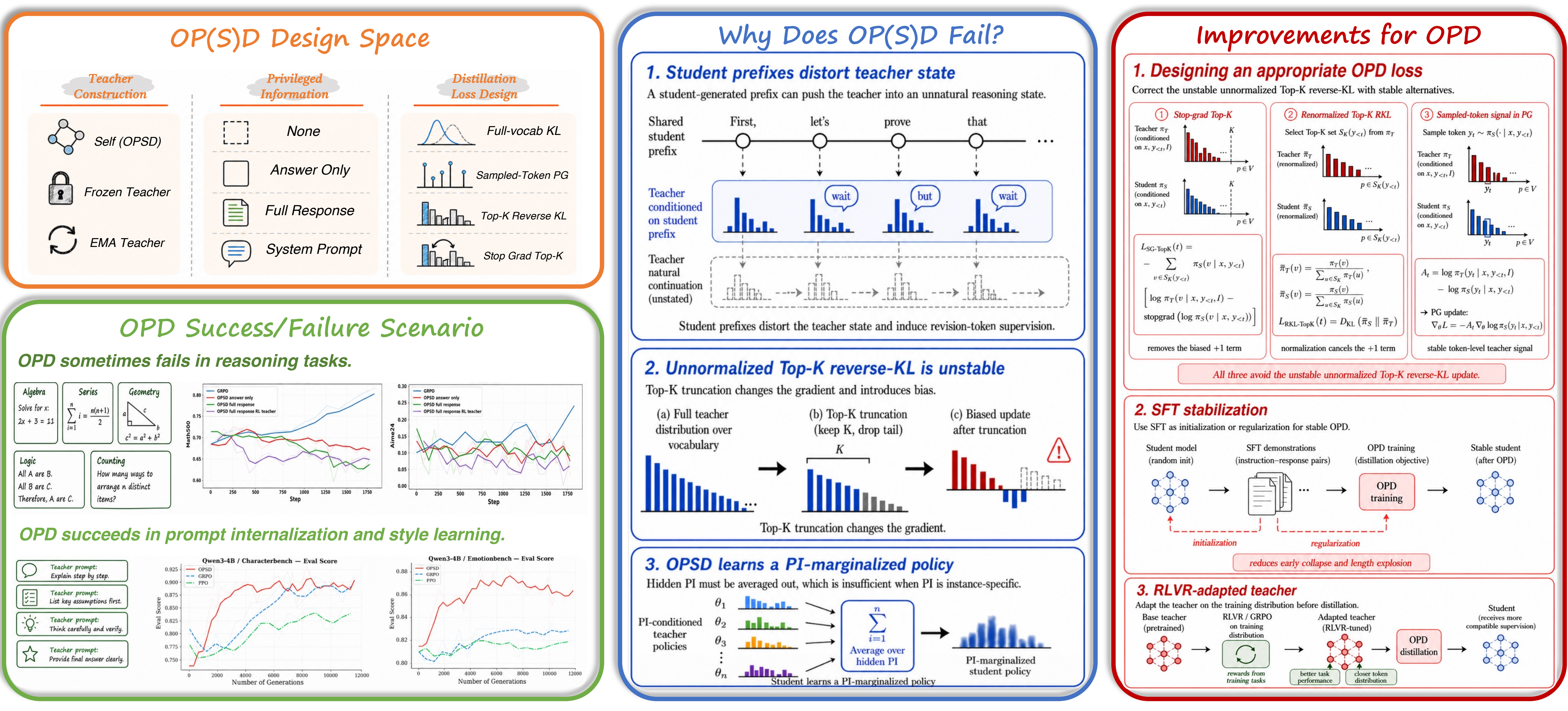

OP(S)D Design Space
Teacher Construction
Privileged Information
Distillation Loss Design
Self (OPSD)
Frozen Teacher
EMA Teacher
None
Answer Only
Full Response
System Prompt
Full-vocab KL
Sampled-Token PG
Top-K Reverse KL
Stop Grad Top-K
OPD Success/Failure Scenario
OPD sometimes fails in reasoning tasks.
Algebra
Solve for x: 2x + 3 = 11
Series
Geometry
Logic
All A are B. All B are C. Therefore, A are C.
Counting
How many ways to arrange n distinct items?
GRPO
OPSD answer only
OPSD full response
OPSD full response RL teacher
Math500
Aime24
Step
OPD succeeds in prompt internalization and style learning.
Teacher prompt: Explain step by step.
Teacher prompt: List key assumptions first.
Teacher prompt: Think carefully and verify.
Teacher prompt: Provide final answer clearly.
Qwen3-4B / Characterbench — Eval Score
Qwen3-4B / Emotionbench — Eval Score
OPSD
GRPO
PPO
Eval Score
Number of Generations
Why Does OP(S)D Fail?
1. Student prefixes distort teacher state
A student-generated prefix can push the teacher into an unnatural reasoning state.
Shared student prefix
First,
let's
prove
that
Teacher conditioned on student prefix
wait
but
wait
Teacher natural continuation (unstated)
Student prefixes distort the teacher state and induce revision-token supervision.
2. Unnormalized Top-K reverse-KL is unstable
Top-K truncation changes the gradient and introduces bias.
(a) Full teacher distribution over vocabulary
(b) Top-K truncation (keep K, drop tail)
K
(c) Biased update after truncation
Top-K truncation changes the gradient.
3. OPSD learns a PI-marginalized policy
Hidden PI must be averaged out, which is insufficient when PI is instance-specific.
PI-conditioned teacher policies
θ_1
θ_2
θ_3
θ_n
$\sum_{i=1}^{n}$
Average over hidden PI
PI-marginalized student policy
Student learns a PI-marginalized policy
Improvements for OPD
1. Designing an appropriate OPD loss
Correct the unstable unnormalized Top-K reverse-KL with stable alternatives.
① Stop-grad Top-K
Teacher π_T (conditioned on $x, y_{<t}, I$)
Student π_S (conditioned on $x, y_{<t}$)
$p \in V$
$L_{\text{SG-TopK}}(t) = -\sum_{v \in S_K(y_{<t})} \pi_S(v \mid x, y_{<t}) \left[\log \pi_T(v \mid x, y_{<t}, I) - \text{stopgrad}\left(\log \pi_S(v \mid x, y_{<t})\right)\right]$
removes the biased +1 term
② Renormalized Top-K RKL
Select Top-K set $S_K(y_{<t})$ from π_T
Teacher $\bar{\pi}_T$ (renormalized)
Student $\bar{\pi}_S$ (renormalized)
$p \in S_K(y_{<t})$
$\bar{\pi}_T(v) = \frac{\pi_T(v)}{\sum_{u \in S_K} \pi_T(u)}$,
$\bar{\pi}_S(v) = \frac{\pi_S(v)}{\sum_{u \in S_K} \pi_S(u)}$
$L_{\text{RKL-TopK}}(t) = D_{\text{KL}}(\bar{\pi}_S \parallel \bar{\pi}_T)$
normalization cancels the +1 term
③ Sampled-token signal in PG
Sample token $y_t \sim \pi_S(\cdot \mid x, y_{<t})$
Teacher π_T (conditioned on $x, y_{<t}, I$)
Student π_S (conditioned on $x, y_{<t}$)
y_t
$A_t = \log \pi_T(y_t \mid x, y_{<t}, I) - \log \pi_S(y_t \mid x, y_{<t})$
→ PG update: $\nabla_\theta L = -A_t \nabla_\theta \log \pi_S(y_t \mid x, y_{<t})$
stable token-level teacher signal
All three avoid the unstable unnormalized Top-K reverse-KL update.
2. SFT stabilization
Use SFT as initialization or regularization for stable OPD.
Student model (random init)
SFT demonstrations (instruction–response pairs)
OPD training (distillation objective)
Stable student (after OPD)
OPD training
initialization
regularization
reduces early collapse and length explosion
3. RLVR-adapted teacher
Adapt the teacher on the training distribution before distillation.
Base teacher (pretrained)
RLVR / GRPO on training distribution
Adapted teacher (RLVR-tuned)
Student (receives more compatible supervision)
rewards from training tasks
better task performance
closer token distribution
OPD distillation